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IdeaDNA

Database Schema Document (SCHEMA)

"Complete Data Model & Relationships"

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1. Database Overview

PostgreSQL 15 with Prisma 5.x ORM. pgvector extension for semantic search. 9 core entities with clear relationships. Designed for scalability with proper indexing and connection pooling.

2. Complete Prisma Schema

```
// Enums enum Mode { GENERATOR EVOLUTION ARCHAEOLOGY } enum Status { DRAFT SELECTED COMPLETED ARCHIVED } enum Platform { WEB MOBILE AI_ML BLOCKCHAIN IOT DESKTOP EXTENSION API_BACKEND } enum SkillLevel { BEGINNER INTERMEDIATE ADVANCED } enum Difficulty { EASY MEDIUM HARD } // Core: User model User { id String @id @default(uuid()) email String @unique name String? skillLevel SkillLevel @default(BEGINNER) budget Int @default(0) teamSize Int @default(1) timeline String @default("2 weeks") techPrefs String[] @default([]) learningGoals String[] @default([]) industry String? yearsOfExperience Int? preferredPlatform Platform? stylePreference String? projects Project[] createdAt DateTime @default(now()) updatedAt DateTime @updatedAt } // Core: Project (main session container) model Project { id String @id @default(uuid()) userId String user User @relation(fields: [userId], references: [id], onDelete: Cascade) title String @default("Untitled Project") description String? mode Mode status Status @default(DRAFT) originalInput String ideas Idea[] selectedIdeaId String? @unique selectedIdea Idea? @relation("SelectedIdea", fields: [selectedIdeaId], references: [id]) blueprint Blueprint? uiMockup UiMockup? evolutionTree Json? archaeologyReport Json? embedding Unsupported("vector(1536)")? parentProjectId String? parentProject Project? @relation("ProjectEvolution", fields: [parentProjectId], references: [id]) childProjects Project[] @relation("ProjectEvolution") createdAt DateTime @default(now()) updatedAt DateTime @updatedAt @@index([userId]) @@index([mode]) @@index([status]) @@index([createdAt]) } // Core: Idea (lightweight card) model Idea { id String @id @default(uuid()) projectId String project Project @relation(fields: [projectId], references: [id], onDelete: Cascade) title String description String problemStatement String solution String keyFeatures String[] innovationScore Int feasibilityScore Int difficulty Difficulty tags String[] platform Platform selectedBy Project? @relation("SelectedIdea") blueprint Blueprint? uiMockup UiMockup? createdAt DateTime @default(now()) @@index([projectId]) @@index([innovationScore]) } // Deep Dive: Blueprint (heavy, lazy-loaded) model Blueprint { id String @id @default(uuid()) ideaId String @unique idea Idea @relation(fields: [ideaId], references: [id], onDelete: Cascade) projectId String @unique project Project @relation(fields: [projectId], references: [id], onDelete: Cascade) overview String @db.Text prd Json techStack Json databaseSchema Json apiDesign Json userFlow Json implementationPlan Json markdown String @db.Text openApiSpec String? @db.Text mermaidDiagrams Json? createdAt DateTime @default(now()) } // Deep Dive: UI Mockup (heavy, lazy-loaded) model UiMockup { id String @id @default(uuid()) ideaId String @unique idea Idea @relation(fields: [ideaId], references: [id], onDelete: Cascade) projectId String @unique project Project @relation(fields: [projectId], references: [id], onDelete: Cascade) html String @db.Text css String? @db.Text components Json styleVariant String @default("MODERN") mobileNotes String? @db.Text tabletNotes String? @db.Text createdAt DateTime @default(now()) } // Curated Database: Pre-loaded ideas model CuratedIdea { id String @id @default(uuid()) title String description String problemStatement String solution String keyFeatures String[] platform Platform subCategory String innovationScore Int marketPotential Int complexity Difficulty suggestedStack Json tags String[] source String trendingScore Int @default(0) embedding Unsupported("vector(1536)")? createdAt DateTime @default(now()) updatedAt DateTime @updatedAt @@index([platform]) @@index([complexity]) @@index([innovationScore]) } // Archaeology: Code analysis reports model ArchaeologyReport { id String @id @default(uuid()) projectId String @unique project Project @relation(fields: [projectId], references: [id], onDelete: Cascade) inputType String inputUrl String? inputCode String? @db.Text codeAnalysis Json patterns Json evolutionTimeline Json thinkingProcess Json scores Json futureVersions Json rawAnalysis String? @db.Text embedding Unsupported("vector(1536)")? createdAt DateTime @default(now()) updatedAt DateTime @updatedAt }
```

3. Entity Relationship Diagram

User (1)	----	<	Project (N)	----	<	Idea (N)
Project (1)	----		(1) Idea	[selectedIdea]		
Idea (1)	----		(1) Blueprint			
Idea (1)	----		(1) UiMockup			
Project (N)	----		(1) Project	[parentProject]		
Project (1)	----		(1) ArchaeologyReport			

4. Indexing Strategy

Table	Index Name	Fields	Type	Purpose
Project	idx_project_user	userId	BTree	User project list queries
Project	idx_project_mode_status	mode, status	BTree	Dashboard filtering
Project	idx_project_created	createdAt	BTree	Sorting and pagination
Idea	idx_idea_project	projectId	BTree	Project ideas query
Idea	idx_idea_score	innovationScore	BTree	Score-based sorting
CuratedIdea	idx_curated_platform	platform	BTree	Platform filtering
CuratedIdea	idx_curated_complexity	complexity	BTree	Complexity filtering
CuratedIdea	idx_curated_embedding	embedding	GIN	Semantic similarity search
Project	idx_project_embedding	embedding	GIN	Similar project search

5. Migration Strategy

- Initial Migration: Create all tables, enums, indexes
- Seed Migration: Insert 500+ curated ideas with embeddings
- Vector Extension: Enable pgvector, create GIN indexes
- Future Migrations: Add columns, create new tables, backfill data

— *End of SCHEMA* —